intel digital readiness

# AI for Workforce: Pre-packaged Courses

# Pre-packaged Courses

Total 8 courses having total content duration of 478 h

### Intro to AI (64 h)

- Basic concepts and applications of artificial intelligence (AI)
- Introduction to AI project cycle
- Focus on issues surrounding AI
- Ethics, Bias, Culture, Regulations, and Professional expectations

### Intro to ML (64 h)

- Introduction to Machine Learning concepts
- Introduction to Python applications
- Data Acquisition and Data Modelling
- Supervised and Unsupervised learning

### Natural Language Processing (64 h)

- Fundamental concepts in NLP and text processing
- Creating a Language Recognition application

### AI for Computer Vision (64 h)

- Basic techniques to process images using OpenCV and Python libraries
- Apply AI in CV for common tasks like Image Classification and Object Detection

### Applied Mathematics for AI (64 h)

- Basic concepts for the role of Mathematics in AI
- Application of Statistics, Linear Algebra, Probability
- Application of topics with reference to their role in the AI Project Cycle.

### Data-Centric AI: Data-first approach to AI(64 h)

- Focus on the data aspect of AI
- Learning how to work with data (statistical, text, and visual)
- Continuous improvements to datasets improve AI solutions
- Integrate and manage data pipelines with MLOps

### Intro to Generative AI (64 h)

- Explore applications of generative AI in various fields
- Develop practical skills in AI project development with the use of different tools
- Examine ethical implications of use of generative AI in various fields

### Data Annotation (30 h)

- Introduction to AI & Domains of AI
- Basics of Data Literacy, Data Collection & Data Ethics
- Exploring Tools & Techniques for Data Annotation

# Pre-Packaged Courses Library

| weeks | Introduction to AI (64h) | Introduction to Machine Learning (64h) | Natural Language Processing (64h) | AI for Computer Vision (64h) |
|---|---|---|---|---|
| 1 | Demystifying & Evolution of AI | Introduction to Machine Learning | AI for Natural Language Processing | Intro to Computer Vision |
| 2 | AI & Technology | Tools to Implement AI (Python & Math) - I | Data Acquisition and Storage for NLP | Mathematics for Computer Vision |
| 3 | Industry 4.0 – Digitalization | Tools to Implement AI (Python & Math) - II | Data Curation NLP Tools and Sources | Data Acquisition for Computer Vision |
| 4 | Domains of AI | No-code Tool for Data Exploration | NLP Data Visualization | Data Exploration - I |
| 5 | AI Project Cycle I | Supervised Learning – I (Modeling) | NLP Data Preprocessing – Part I | Data Exploration - II |
| 6 | AI Project Cycle II | Supervised Learning – II (Modeling) | NLP Data Preprocessing – Part II | Introduction to CNN |
| 7 | Societal Impact of AI | Unsupervised Learning – I (Modeling) | NLP Data Pipeline | Building Blocks of CNN |
| 8 | Elements of Machine Learning | Unsupervised Learning – II (Modeling) | Intro to Machine Learning Models | Transfer Learning |
| 9 | Elements of Deep Learning | Reinforcement Learning | Neural Language Models | Hands-on with OpenCV |
| 10 | Data – Fuel for AI | Neural Networks | NLP Model Deployment | Hands-on with YOLO |
| 11 | AI/Data Science Team | Deep Learning - I | Building a Chatbot I | Hands-on with TensorFlow & Keras |
| 12 | Tools to Implement AI | Deep Learning - II | Building a Chatbot II (Testing & Deployment) | GAN (Generative Adversarial Network) |
| 13 | AI Project I – Statistical Data | ML Project Supervised Learning AI Models | NLP for Multiple Languages | Intro to OpenVINO toolkit for CV application |
| 14 | AI Project II – Natural Language Processing | ML Project Unsupervised Learning AI Models | LSTM, Transformers and BERT | Explainable AI for Computer Vision |
| 15 | AI Project III – Computer Vision | Deep Learning Project | Advanced NLP Models I | Deploying Computer Vision Applications |
| 16 | Future Possibilities of AI | Future of ML & Deep Learning | Advanced NLP Models II | Future of Computer Vision |

# Pre-Packaged Courses Library

| Weeks | Applied Mathematics for AI (64h) | Data-Centric AI: Data First Approach to AI (64h) | Introduction to Generative AI (64h) |
|---|---|---|---|
| 1 | Intro to Math for AI | Data-Centric AI vs Model-Centric AI | Introduction to Generative AI |
| 2 | Data Acquisition | AI Project Cycle & MLOps | Demystifying Generative AI Models |
| 3 | Descriptive Statistics - I | Fundamentals of Data | Prompt Engineering – I |
| 4 | Descriptive Statistics - II | Data Engineering Pipelines: Fundamentals & Relationship with MLOps | Prompting Engineering - II |
| 5 | Inferential Statistics - I | Data Collection | Basics of Text and Image Generation and Using Gen AI for Productivity - I |
| 6 | Inferential Statistics - II | Data Wrangling (Cleaning) | Basics of Video and Audio Generation and Using Gen AI for Productivity - II |
| 7 | Data Visualization | Data Enrichment | Generative AI Ethics |
| 8 | Representing Data through Dashboards | Data Labeling | Gen AI Project Lifecycle and Generative AI Tools for Development |
| 9 | Linear Algebra - I | Data Exploration | Neural Network Fundamentals |
| 10 | Linear Algebra - II | Modeling & Evaluation | Transformers and the Evolution of Large Language Models |
| 11 | Linear Algebra - III | Introduction to MLOps Platform | Diffusion Models and the Evolution of Text-to-Image Generation |
| 12 | Probability - I | Introduction to AutoML | Pre-Training Generative AI Models |
| 13 | Probability - II | Implementing AutoML 1: Model Building, EDA and Advanced Pre-processing | Fine-Tuning Generative AI Models |
| 14 | Evaluation Metrics | Implementing AutoML 2: Production Deployment, Registration & versioning of Models | RLHF, Evaluating, and Optimizing Generative AI Models |
| 15 | Optimization | Explainable AI & Model Interpretability | Prototyping Generative AI Applications |
| 16 | Hands-on Project | Overview of Containerization & Deployment | LLM Alignment and the Responsible Use of Generative AI |

| Weeks | Data Annotation (30 hours) |
|---|---|
| 1 | Introduction to AI |
| 2 | AI Domains |
| 3 | Basics of Data Literacy |
| 4 | Data Collection |
| 5 | Data Ethics |
| 6 | Processing & Interpreting Data |
| 7 | Data Annotation |